

ON ASYMPTOTIC VARIANCES OF UNIQUENESS
ESTIMATORS IN FACTOR ANALYSIS

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estimator.

Summary: It is shown that the asymptotic variances of
uniqueness estimators in factor analysis
decrease as new observed variates are added to
the model while the number of factors is held
fixed. The limit form of the associated
asymptotic covariance matrix is calculated.

1. INTRODUCTION

In the factor analysis model the $p \times p$ population covariance
matrix Σ is representable in the form

$$(1) \quad \Sigma = \mathbf{A}\mathbf{A}' + \mathbf{\Psi},$$

where $\mathbf{A}$ is a $p \times k$ matrix of factor loadings and $\mathbf{\Psi}$ is a $p \times p$
nonnegative definite diagonal matrix of unique variances (e.g.
Lawley and Maxwell, 1971). Let $\mathbf{S}$ be the sample covariance
matrix giving an unbiased estimator of Σ . Then the parameter
estimators $\hat{\mathbf{A}}$ and $\hat{\mathbf{\Psi}}$ associated with the model (1), are usually

obtained by minimizing a real-valued function $F(S, Z)$ which gives an indication of the discrepancy between S and Z . That is $\hat{A}$ and $\hat{\Psi}$ are calculated as solutions of the minimization problem

$$\underset{A, \Psi}{\text{minimize}} F(S, A + \Psi) .$$

Two standard discrepancy functions employed in computer packages are

$$F_{HL}(S, Z) = \log|Z| - \log|S| + \text{tr}(SZ^{-1}) - p,$$

which corresponds to the method of maximum likelihood under normality assumptions, and a generalized least squares discrepancy function

$$F_{GLS}(S, Z) = \frac{1}{2} \text{tr}[(S - Z)S^{-1}]^2 .$$

It is known that under mild regularity conditions the minimum discrepancy estimators are consistent (Kano 1983, 1986; Shapiro, 1983) and asymptotically normal (Browne, 1982; Shapiro, 1983).

It was observed empirically that the maximum likelihood (generalized least squares) estimator $\hat{\Psi}$ is stable and improper solutions rarely occur when the number of items p is large relative to the number of factors k (e.g. Anderson and Gerbing, 1984). The aim of this note is to show analytically that the asymptotic variances of the uniqueness estimators, (diagonal elements of $\hat{\Psi}$), decrease as new observed variates are added to the model while the number of factors k is held fixed. This gives a theoretical explanation for the above-mentioned empirical observation.

2. A RELATION BETWEEN THE ASYMPTOTIC VARIANCES AND THE NUMBER OF VARIATES

Consider the factor analysis model (1) and let

$$(2) \quad \Sigma_{11} = A_1 A_1' + \Psi_{11}$$

be the $q \times q$ submodel of (1). That is Σ_{11} and Ψ_{11} are the $q \times q$ principal submatrices of Σ and Ψ , respectively, and A_1 is the $q \times k$ upper submatrix of the $p \times k$ matrix A . It can be said that model (2) is obtained from (1) by deleting $p-q$ observed variates while keeping the number of factors fixed. Let Π_p and Π_q be the $p \times p$ and $q \times q$ asymptotic covariance matrices of the uniqueness estimators associated with the models (1) and (2), respectively. It will be assumed that the population distribution is normal and that a normal theory discrepancy function, for example F_{HL} or F_{GLS} , is employed. We denote by $(\Pi_p)_{11}$ the $q \times q$ principal submatrix of the matrix Π_p . Then the following result holds.

Theorem 1. Under the above specifications

$$(3) \quad \Pi_q \geq (\Pi_p)_{11}$$

in the Loewner sense of inequality, i.e. $\Pi_q - (\Pi_p)_{11}$ is a nonnegative definite matrix.

Proof. Let E be a $p \times (p-k)$ matrix representing an orthogonal complement of A , that is $E'A = 0$ and $\text{rank}(E) = p - \text{rank}(A) = p - k$. Then the asymptotic covariance matrix Π_p can be expressed in the form (cf. Lawley and Maxwell, 1971; Jöreskog and Goldberger, 1972; Shapiro, 1983)

$$\Pi_p = 2(Q * Q)^{-1},$$

where

$$(3) \quad Q = E(E' \Sigma E)^{-1} E' = E(E' \Psi E)^{-1} E'$$

and $*$ denotes the term by term (Hadamard) product of two matrices. (For a thorough discussion of the Hadamard product and its properties the reader is referred to Styan (1973).)

Similarly

$$R_Q = 2(R * R)^{-1},$$

where

$$R = F(F' \Psi_{II} F)^{-1} F'$$

with F being an orthogonal complement of A_1 . Let us consider the $p \times p$ matrix

$$\bar{R} = \begin{bmatrix} R & \vdots & 0 \\ \cdot & \cdot & \cdot \\ 0 & \vdots & 0 \end{bmatrix}$$

and show that

$$(4) \quad Q * Q \geq \bar{R} * \bar{R}.$$

Given F , the orthogonal complement E can be constructed in the form $E = [E_1, E_2]$ with $E_1 = [F', 0]'$ and

$$E_1' \Psi E_2 = 0.$$

Now we have from (3) that

$$Q = E_1(E_1' \Psi E_1)^{-1} E_1' + E_2(E_2' \Psi E_2)^{-1} E_2' = \bar{R} + Z,$$

where $Z = E_2(E_2' \Psi E_2)^{-1} E_2'$ is a nonnegative definite matrix.

Then

$$Q * Q = \bar{R} * \bar{R} + 2\bar{R} * Z + Z * Z$$

and, since the Hadamard product of two nonnegative definite

matrices is a nonnegative definite matrix, the inequality (4) follows. Consequently

$$(5) \quad Q * Q + \epsilon I_p \geq \bar{R} * \bar{R} + \epsilon I_p,$$

where I_p denotes the $p \times p$ identity matrix. Since for every $\epsilon > 0$ the corresponding matrices in both sides of (5) are positive definite, it follows that

$$(6) \quad (\bar{R} * \bar{R} + \epsilon I_p)^{-1} \geq (Q * Q + \epsilon I_p)^{-1}.$$

Moreover,

$$\lim_{\epsilon \rightarrow 0^+} [(\bar{R} * \bar{R} + \epsilon I_p)^{-1}]_{11} = (R * R)^{-1}$$

and

$$\lim_{\epsilon \rightarrow 0^+} [(Q * Q + \epsilon I_p)^{-1}]_{11} = [(Q * Q)^{-1}]_{11}.$$

Together with (6) the above relations imply that

$$(R * R)^{-1} \geq [(Q * Q)^{-1}]_{11},$$

which completes the proof. ■

It follows from the result of theorem 1 that the diagonal elements of Σ_q are greater than or equal to the corresponding diagonal elements of Σ_p . This shows that uniqueness estimators under the model (1) are more "precise" than the corresponding estimators under the submodel (2). It will be of interest to evaluate the limit form of the asymptotic variances of uniqueness estimators as the number of observed variates tends to infinity while the number of factors is fixed.

Let us now consider a *sequence* of factor analysis models (1),

$$(7) \quad \Sigma_p = A_p A_p' + \Psi_p,$$

$p = q, q+1, \dots$, with a fixed number of factors k . It will be assumed that for every p , $p \geq q$, the model (2) is the corresponding submodel of (7) and that (cf. Williams, 1978),

$$(8) \quad \lim_{p \rightarrow \infty} (A_p' \Psi_p^{-1} A_p)^{-1} = 0.$$

We shall keep essentially the same notation as in theorem 1. In particular for a given p , Π_p denotes the $p \times p$ asymptotic covariance matrix associated with (7) and $(\Pi_p)_{11}$ denotes the $q \times q$ principal submatrix of Π_p .

Theorem 2. Under the above specifications and condition (8),

$$(9) \quad \lim_{p \rightarrow \infty} (\Pi_p)_{11} = 2\Psi_q^{-2}.$$

Proof. We have that

$$\Pi_p = 2(Q_p * Q_p)^{-1},$$

where matrix Q_p can be expressed in the form

$$Q_p = \Psi_p^{-1} - \Psi_p^{-1} A_p (A_p' \Psi_p^{-1} A_p)^{-1} A_p' \Psi_p^{-1},$$

which is equivalent to formula (3) (see, e.g., Rao, 1973, p.77).

Then

$$Q_p = \Psi_p^{-1/2} (I_p - M_p) \Psi_p^{-1/2},$$

where

$$M_p = \Psi_p^{-1/2} A_p (A_p' \Psi_p^{-1} A_p)^{-1} A_p' \Psi_p^{-1/2}.$$

Notice that because of the condition (8) we have

$$(10) \quad \lim_{p \rightarrow \infty} (M_p)_{11} = \lim_{p \rightarrow \infty} \Psi_q^{-1/2} A_q (A_q' \Psi_q^{-1} A_q)^{-1} A_q' \Psi_q^{-1/2} = 0.$$

Also, the matrix M_p and then $M_p * M_p$ are nonnegative definite and hence $M_p * M_p$ can be represented in the form

$$M_p * M_p = B_p B_p'.$$

After some calculations we obtain

$$Q_p * Q_p = \Psi_p^{-1} (D_p + B_p B_p') \Psi_p^{-1},$$

where $D_p = I_p - 2 \operatorname{diag}(M_p)$.

It follows that

$$(Q_p * Q_p)^{-1} = \Psi_p (D_p^{-1} - D_p^{-1} B_p (I_p + B_p D_p^{-1} B_p)^{-1} B_p D_p^{-1}) \Psi_p.$$

Let D_{11} be the $q \times q$ principal submatrix of the diagonal matrix D_p and B_1 be the corresponding submatrix of B_p . Then since D_p is diagonal

$$(11) \quad [(Q_p * Q_p)^{-1}]_{11} = \Psi_q [D_{11}^{-1} - D_{11}^{-1} B_1 (I_p + B_p D_p^{-1} B_p)^{-1} B_1 D_{11}^{-1}] \Psi_q.$$

It follows from (10) that

$$(12) \quad \lim_{p \rightarrow \infty} D_{11} = \lim_{p \rightarrow \infty} \{I_q - 2 \operatorname{diag}(M_p)_{11}\} = I_q.$$

Moreover,

$$B_1 (I_p + B_p D_p^{-1} B_p)^{-1} B_1' \leq B_1 B_1'$$

and hence

$$\|B_1 (I_p + B_p D_p^{-1} B_p)^{-1} B_1'\| \leq \|B_1 B_1'\|,$$

where $\|A\|$ denotes the largest eigenvalue of a nonnegative definite symmetric matrix A . ($\|A\|$ is the operator norm of A corresponding to the Euclidean norm.) Since $B_1 B_1' = (M_p)_{11}$ and (10), this implies that

$$(13) \quad \lim_{p \rightarrow \infty} B_1 (I_p + B_p D_p^{-1} B_p)^{-1} B_1' = 0.$$

Finally

$$(\Sigma_p)_{11} = 2[(Q_p * Q_p)^{-1}]_{11}$$

and hence (9) follows from (11), (12) and (13).

The results of theorems 1 and 2 show that as the number of observed variates increases the asymptotic covariance matrix of, say the first q , uniqueness estimators decreases and under condition (8) approaches the diagonal matrix $2\sigma_q^2$.

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